

6. API Endpoints Tested

Overview

API endpoints are specific URLs that allow communication between the client application and the server. Each endpoint performs a particular operation such as retrieving data, creating records, updating information, or deleting records.

In this project, multiple endpoints of the Fake Store REST API were tested to validate the functionality of different modules including Authentication, Products, Categories, Cart, and Users.

The testing covered various HTTP methods such as GET, POST, PUT, PATCH, and DELETE to ensure that the APIs work correctly and return expected responses.

The base URL used throughout the project is:

`https://fakestoreapi.com`

6.1 Authentication API

Purpose

The Authentication API is used to verify user login functionality and generate an authentication token.

The generated token can be used for authorized operations in real-world applications.

API Name

Login User

Endpoint

`POST /auth/login`

Complete URL

`https://fakestoreapi.com/auth/login`

Request Body

```
{
  "username": "mor_2314",
```

```
"password": "83r5^_"
}
```

Expected Response

```
{
  "token": "generated_token"
}
```

Validations Performed

- Verify Status Code is 200.
 - Verify token is generated.
 - Verify response body is not empty.
 - Verify response time is within acceptable limits.
 - Verify JSON response structure.
-

6.2 Products API

Purpose

The Products API is used to manage product-related operations in the e-commerce application.

It allows users to retrieve, create, update, and delete product information.

API 1: Get All Products

Endpoint

GET /products

Complete URL

<https://fakestoreapi.com/products>

Purpose

Retrieve all available products.

Validations Performed

- Status Code Validation
- Response Time Validation

- Product List Validation
 - JSON Response Validation
-

API 2: Get Single Product

Endpoint

GET /products/1

Complete URL

https://fakestoreapi.com/products/1

Purpose

Retrieve details of a specific product using Product ID.

Validations Performed

- Verify Product ID
 - Verify Product Title
 - Verify Product Price
 - Verify Status Code
 - Verify Response Time
-

API 3: Add New Product

Endpoint

POST /products

Complete URL

https://fakestoreapi.com/products

Sample Request Body

```
{
  "title": "Laptop",
  "price": 45000,
  "description": "Gaming Laptop",
  "image": "https://i.pravatar.cc",
  "category": "electronics"
}
```

Purpose

Create a new product record.

Validations Performed

- Verify Status Code
 - Verify Product Creation
 - Verify Response Body
 - Verify Product Details Returned
-

API 4: Update Product (PUT)

Endpoint

PUT /products/1

Complete URL

<https://fakestoreapi.com/products/1>

Purpose

Replace all product details with new data.

Validations Performed

- Verify Status Code
 - Verify Updated Product Details
 - Verify Response Structure
 - Verify Response Time
-

API 5: Update Product (PATCH)

Endpoint

PATCH /products/1

Complete URL

<https://fakestoreapi.com/products/1>

Purpose

Update only selected fields of a product.

Validations Performed

- Verify Status Code
 - Verify Updated Fields
 - Verify Response Data
 - Verify Response Time
-

API 6: Delete Product

Endpoint

DELETE /products/1

Complete URL

https://fakestoreapi.com/products/1

Purpose

Delete an existing product.

Validations Performed

- Verify Status Code
 - Verify Product Deletion Response
 - Verify Response Body
 - Verify Response Time
-

6.3 Categories API

Purpose

The Categories API is used to retrieve category information and products belonging to specific categories.

API 1: Get All Categories

Endpoint

GET /products/categories

Complete URL

`https://fakestoreapi.com/products/categories`

Purpose

Retrieve all available product categories.

Validations Performed

- Verify Status Code
 - Verify Category List
 - Verify Response Structure
 - Verify Response Time
-

API 2: Get Products in Category

Endpoint

`GET /products/category/{categoryName}`

Example

`GET /products/category/electronics`

Complete URL

`https://fakestoreapi.com/products/category/electronics`

Purpose

Retrieve all products belonging to a specific category.

Validations Performed

- Verify Status Code
 - Verify Product List
 - Verify Category Name
 - Verify Response Structure
-

6.4 Cart API

Purpose

The Cart API is used to manage shopping cart operations within the e-commerce application.

API 1: Get All Carts

Endpoint

GET /carts

Complete URL

https://fakestoreapi.com/carts

Purpose

Retrieve all cart records.

Validations Performed

- Verify Status Code
 - Verify Cart Data
 - Verify Response Structure
 - Verify Response Time
-

API 2: Get Single Cart

Endpoint

GET /carts/1

Complete URL

https://fakestoreapi.com/carts/1

Purpose

Retrieve a specific cart using Cart ID.

Validations Performed

- Verify Cart ID
- Verify Product Information
- Verify Status Code
- Verify Response Time

API 3: Add to Cart

Endpoint

POST /carts

Complete URL

<https://fakestoreapi.com/carts>

Purpose

Add products to a shopping cart.

Validations Performed

- Verify Status Code
- Verify Cart Creation
- Verify Response Data
- Verify Response Structure

API 4: Update Cart (PUT)

Endpoint

PUT /carts/1

Purpose

Replace the entire cart information with new data.

Validations Performed

- Verify Status Code
- Verify Updated Cart Data
- Verify Response Structure

API 5: Update Cart (PATCH)

Endpoint

PATCH /carts/1

Purpose

Partially update cart information.

Validations Performed

- Verify Status Code
 - Verify Updated Fields
 - Verify Response Data
-

API 6: Delete Cart

Endpoint

`DELETE /carts/1`

Purpose

Delete a cart record.

Validations Performed

- Verify Status Code
 - Verify Delete Response
 - Verify Response Time
-

6.5 Users API

Purpose

The Users API is used to retrieve user information stored in the application.

API 1: Get All Users

Endpoint

`GET /users`

Complete URL

`https://fakestoreapi.com/users`

Purpose

Retrieve all registered users.

Validations Performed

- Verify Status Code
 - Verify User List
 - Verify Response Structure
 - Verify Response Time
-

API 2: Get Single User

Endpoint

GET /users/1

Complete URL

https://fakestoreapi.com/users/1

Purpose

Retrieve details of a specific user.

Validations Performed

- Verify User ID
 - Verify User Details
 - Verify Status Code
 - Verify Response Structure
-

Summary

A total of **17 API endpoints** were tested across five major modules: Authentication, Products, Categories, Cart, and Users. Each endpoint was validated for functionality, status code accuracy, response data correctness, response structure, and response time. The testing ensured that all APIs behaved according to expected business requirements and provided reliable communication between the client and server.